On the Existence of Dynamics

Relational Closure and Stable Iterability as Operational Principles

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Abstract

We ask for minimal conditions under which a relation can exist as a dynamical object. Our starting point is a single structural principle: a relation that is not closed under its own admissible composition cannot exist as a dynamical object. We formulate this principle operationally in terms of *embeddability*: a candidate relation is dynamical only if it can be embedded into a compositionally consistent iteration family, with endpoint-defined iterates independent of decomposition scheme.

When sufficiently small increments exist, embeddability admits a local representation in terms of an induced generator, but the generator is treated as derived rather than primitive. To distinguish mere iterability from viable local dynamics, we impose a minimal stability condition:

$$C(\varepsilon)^T C(\varepsilon) = I.$$

Its infinitesimal consequence is

$$\Gamma^T + \Gamma = 0,$$

so the induced local generator is forced into the skew-symmetric sector. In this sense, the Euclidean local sector is not postulated independently, but emerges as the canonical infinitesimal realization of the minimal stability requirement.

This has a sharp consequence: nontrivial stable local dynamics cannot begin in the real scalar sector. In the corresponding local canonical form, stable generators decompose into two-dimensional rotation blocks generated by an operator J satisfying $J^2 = -I$. Thus the first admissible nontrivial stable local mode is necessarily two-dimensional and rotational. That block carries a natural complex structure. In the associated one-dimensional complex representation, the canonical generator J is represented by multiplication by i, so the full local generator ωJ is represented by multiplication by $i\omega$ for some real ω. In this sense, the imaginary unit appears not as a prior postulate, but as the scalar realization of the first nontrivial stable local mode.

The resulting framework isolates dynamics as a compositional phenomenon: iterable relational structure is primary, while local generators and parameterizations arise only as secondary representations.

1 Introduction

Dynamics is usually introduced by postulating a parameter and a law of change. In that order of presentation, the existence of dynamics is assumed before it is examined. In the present work we reverse this order and ask a prior question: *under what conditions can a relation exist as a dynamical object at all?*

Our starting point is that a relation should not count as dynamical merely because one can write an update rule for it. If repeated continuation destroys the class to which the relation belongs, or if the resulting iterate depends on how the continuation is organized, then no endpoint-defined object of evolution has been specified. What remains is at best a procedure, not a dynamical object.

To make this precise, we formulate an operational notion of *embeddability*. A candidate relation is dynamical only if it can be embedded into a compositionally consistent iteration family

$$\varepsilon \mapsto C(\varepsilon),$$

with identity at $\varepsilon = 0$, closure under composition, and decomposition invariance. The parameter ε is not initially interpreted as physical time. It is only an iteration parameter indexing successive admissible continuations of the same relational object.

In this framework, a local generator is not postulated at the outset. It appears only if the iteration family already exists and is sufficiently regular near the identity. Thus the generator is treated as a derived local representation of an already existing iterable relation, rather than as the primitive origin of the dynamics.

To distinguish mere iterability from viable local dynamics, we impose a minimal stability condition:

$$C(\varepsilon)^T C(\varepsilon) = I.$$

Its infinitesimal consequence is

$$\Gamma^T + \Gamma = 0,$$

so the induced local generator is forced into the skew-symmetric sector. This leads to a canonical local rotational structure and shows that the real one-dimensional stable local sector is trivial, since in dimension one skew-symmetry forces the local generator to vanish. The first nontrivial stable local mode therefore appears in a two-dimensional rotational block. That canonical block carries a natural complex structure. In the associated one-dimensional complex representation, the canonical generator J is represented by multiplication by i, so the full local generator ωJ is represented by multiplication by $i\omega$ for some real ω.

The central result of the paper is therefore the following: nontrivial stable local dynamics cannot begin in the real scalar sector. Its first admissible nontrivial stable local realization is necessarily two-dimensional, rotational, and canonically complex.

The resulting picture is principle-based: dynamics is fundamentally compositional, while local generators and parameter-based descriptions arise only as secondary representations of an already defined iterable relational object.

2 Minimal setup: relations and composition

2.1 Labels and relations

Let $\mathcal{X}$ be a nonempty set of labels. At this stage, the labels carry no additional structure: no ordering, topology, metric, or background geometry is assumed. They serve only as endpoints for relational composition.

A relation from $a \in \mathcal{X}$ to $b \in \mathcal{X}$ is denoted by

$$C(b, a).$$

The ordering is essential: $C(b, a)$ is the relation from a to b, not vice versa.

2.2 Composition

Whenever endpoints match, relations may be composed. Given $a, b, c \in \mathcal{X}$, the composite of $C(b, a)$ and $C(c, b)$ is written as

$$C(c, b)\, C(b, a),$$

and is interpreted as the relation from a to c.

We assume associativity of composition whenever all products are defined. Thus, for compatible endpoints,

$$C(d, c)\big(C(c, b)C(b, a)\big) = \big(C(d, c)C(c, b)\big)C(b, a).$$

For each endpoint $a \in \mathcal{X}$, we also assume an identity relation

$$C(a, a) = I,$$

satisfying the usual unit property whenever the products are defined.

2.3 Reversibility

In the present paper we restrict attention to the reversible case. Thus every admissible relation is assumed to have an inverse,

$$C(b, a)^{-1} = C(a, b),$$

with

$$C(b, a)\, C(a, b) = I, \qquad C(a, b)\, C(b, a) = I.$$

This assumption is made for structural clarity: the existence problem studied here is already nontrivial in the reversible setting, and no irreversible sector is needed for the present argument.

2.4 Admissible class

Not every formally writable relation is treated as dynamically admissible. We therefore distinguish an admissible class

$$\mathcal{K}$$

of relations. The central question of the paper is not whether a single relation can be written, but under what conditions a relation in $\mathcal{K}$ can qualify as a genuine dynamical object.

At this stage, $\mathcal{K}$ is left abstract. No linear structure, generator, or local law is assumed. In particular, the present paper does not assume a priori that $\mathcal{K}$ is linear or globally represented in coordinates. The only primitive issue is whether admissible relations can support coherent continuation within one and the same class. Only later, near the identity and in a chosen local representation, will additional regularity be imposed.

2.5 Composition versus iteration

The distinction between composition and iteration is crucial. Composition refers to the concatenation of relations with matching endpoints, for example

$$C(c, b)\, C(b, a).$$

Iteration, by contrast, will later refer to repeated application of one and the same admissible relation through an associated family

$$\varepsilon \mapsto C(\varepsilon).$$

Thus Section 2 fixes only the endpoint-based kinematical setting. The operational question is therefore not yet how a relation changes, but when a single admissible relation can support coherent self-continuation as one object.

3 Closure principle

We now state the single principle on which the rest of the paper is based. It is meant as a criterion of relational dynamical objecthood, not as a specific law of motion.

Axiom 1 (C0: Relational Closure)**.** *Relations that are not closed under their own admissible composition cannot exist as dynamical objects.*

A change that is not relationally comparable cannot yet count as dynamical. A merely formal update, taken by itself, does not yet determine a dynamical object. Only when admissible continuation remains comparable within one coherent class can one speak of the continued existence of the same relational object. In this sense, closure is not an additional technical requirement imposed on dynamics from outside; it expresses a prior condition under which relational change can count as dynamical.

The content of Axiom 1 is structural. A relation qualifies as a dynamical object only if its admissible continuation remains within one coherent class. If repeated admissible composition necessarily drives the relation outside that class, then no well-defined object of evolution has been specified. What remains may still be a rule of update or a formal prescription, but not a dynamics in the sense considered here.

At this stage, the principle is intentionally minimal. It does not assume a preferred parameter, infinitesimal generator, local smoothness, or geometric background. Its role is only to state a necessary condition of existence: before asking how a relation evolves, one must first ask whether it can survive its own admissible continuation as an object of one coherent kind.

The next task is therefore operational. We must translate closure under admissible self-composition into concrete conditions that can be checked and used. This leads to the notion of embeddability.

4 Operational formulation: embeddability

The closure principle must now be translated into an operational condition. The central question is the following: when can a single admissible relation be continued coherently as one and the same relational object?

4.1 Embeddability

Let $\mathcal{K}$ be the admissible class of relations, and let $C \in \mathcal{K}$ be a single candidate relation. We now formulate the operational condition under which C qualifies as dynamically iterable. Namely, there must exist a one-parameter family

$$\varepsilon \mapsto C(\varepsilon) \in \mathcal{K}$$

such that

$$C(0) = I, \qquad C(\varepsilon_1)C(\varepsilon_2) = C(\varepsilon_1 + \varepsilon_2), \tag{1}$$

whenever the compositions are admissible, and such that $C(\varepsilon_0) = C$ for some $\varepsilon_0 \neq 0$.

Here the identity I plays the role of zero relational deviation, that is, the local reference point against which coherent continuation is tested. The parameter ε is not yet interpreted as physical time. It is only an iteration parameter indexing successive admissible continuations of the same relation. Equation (1) expresses that iteration remains internal to one coherent family.

Definition 1 (Embeddability)**.** *A relation $C \in \mathcal{K}$ is called* embeddable *if it admits a family $C(\varepsilon) \in \mathcal{K}$ satisfying*

$$C(0) = I, \qquad C(\varepsilon_1)C(\varepsilon_2) = C(\varepsilon_1 + \varepsilon_2),$$

with $C(\varepsilon_0) = C$ for some $\varepsilon_0 \neq 0$.

For the present argument, the semigroup formulation is sufficient. In the reversible setting, one may locally extend it to a group formulation whenever negative increments are also admitted.

4.2 Decomposition invariance

Embeddability is not only closure under repeated composition. It also requires that the iterate associated with a total increment be independent of how that increment is decomposed.

Thus, if

$$\varepsilon = \varepsilon_1 + \cdots + \varepsilon_n,$$

then one must have

$$C(\varepsilon) = C(\varepsilon_n) \cdots C(\varepsilon_2) C(\varepsilon_1), \tag{2}$$

with the right-hand side representing the same iterate $C(\varepsilon)$, not a scheme-dependent construction. Refinement changes only the description of the iterate, not the relational object itself.

In this sense, embeddability is the operational realization of the closure principle: a relation exists dynamically only if it admits a compositionally consistent and decomposition-independent iteration family.

4.3 Local regularity and induced generators

Embeddability alone does not yet imply a local differential description. The composition law

$$C(\varepsilon_1 + \varepsilon_2) = C(\varepsilon_2) C(\varepsilon_1)$$

states only that the iterable family is compositionally consistent. To pass from this global operational structure to a local generator, one must impose an additional regularity assumption near $\varepsilon = 0$, for example continuity or differentiability in a chosen local representation near the identity.

Assume therefore that $C(\varepsilon)$ is differentiable at $\varepsilon = 0$. Then the first-order behavior of the family near the identity is well defined, and one may introduce the induced local generator by

$$\Gamma := \left. \frac{d}{d\varepsilon} C(\varepsilon) \right|_{\varepsilon=0}. \tag{3}$$

Equivalently, the iterable family admits the first-order local expansion

$$C(\varepsilon) = I + \varepsilon\, \Gamma + o(\varepsilon) \qquad (\varepsilon \to 0). \tag{4}$$

Thus the generator is not postulated independently. It is extracted from the already existing iterable family as its first-order local representative near the identity.

If, in addition, the composition law holds for nearby increments and the iterable family is locally regular, then the local differential form of the iteration follows from the composition law together with the first-order expansion at the identity. Indeed, writing

$$C(\varepsilon + h) = C(h) C(\varepsilon)$$

and using the local expansion

$$C(h) = I + h\, \Gamma + o(h) \qquad (h \to 0),$$

one obtains

$$C(\varepsilon + h) = \big(I + h\, \Gamma + o(h)\big) C(\varepsilon) = C(\varepsilon) + h\, \Gamma C(\varepsilon) + o(h).$$

Therefore

$$\frac{C(\varepsilon + h) - C(\varepsilon)}{h} = \Gamma C(\varepsilon) + \frac{o(h)}{h},$$

and passing to the limit $h \to 0$ yields

$$\frac{d}{d\varepsilon} C(\varepsilon) = \Gamma\, C(\varepsilon), \qquad C(0) = I.$$

In the constant-generator case, the solution is

$$C(\varepsilon) = \exp(\varepsilon\,\Gamma). \tag{5}$$

Hence the exponential representation is not an independent postulate. It is the local realization of a compositionally consistent one-parameter family once sufficient regularity near the identity is assumed.

Remark 1. The logical order is therefore:

$$\text{closure principle} \longrightarrow \text{embeddability} \longrightarrow \text{decomposition invariance} \longrightarrow \text{induced generator.}$$

The exponential form appears only at the final stage, and only under the additional assumption of local regularity. The standard generator-first viewpoint is thus reversed.

5 Operational formulation: stability

Embeddability expresses the existence of a relational object under self-composition. Yet embeddability alone guarantees only compositional consistency; it does not exclude locally amplifying or collapsing modes. We now pass to a real local linear representation of the iterable family, in which transpose is defined, and impose the minimal stability condition:

$$C(\varepsilon)^T C(\varepsilon) = I. \tag{6}$$

This is a local sector choice: at this stage we ask for the minimal real local condition under which continuation preserves size rather than producing local amplification or collapse. It is not introduced by assuming a Euclidean geometry in advance; rather, the Euclidean local sector emerges as the canonical infinitesimal realization of this stability condition.

5.1 Infinitesimal consequence

Assume that the iteration family admits a first-order local representation near $\varepsilon = 0$,

$$C(\varepsilon) = I + \varepsilon\,\Gamma + o(\varepsilon).$$

In the locally constant-generator case this is equivalent to

$$C(\varepsilon) = \exp(\varepsilon\,\Gamma).$$

Substituting this into (6) gives

$$\big(I + \varepsilon\,\Gamma^T + o(\varepsilon)\big)\big(I + \varepsilon\,\Gamma + o(\varepsilon)\big) = I.$$

Expanding to first order in ε, one obtains

$$I + \varepsilon(\Gamma^T + \Gamma) + o(\varepsilon) = I,$$

hence

$$\Gamma^T + \Gamma = 0. \tag{7}$$

Thus the induced local generator is forced into the skew-symmetric sector.

Proposition 1 (Infinitesimal stability)**.** *If the iteration family $C(\varepsilon)$ is locally regular and satisfies*

$$C(\varepsilon)^T C(\varepsilon) = I,$$

then its induced local generator satisfies

$$\Gamma^T + \Gamma = 0.$$

Proof. Write

$$C(\varepsilon) = I + \varepsilon\Gamma + o(\varepsilon).$$

Then

$$C(\varepsilon)^T C(\varepsilon) = \big(I + \varepsilon\Gamma^T + o(\varepsilon)\big)\big(I + \varepsilon\Gamma + o(\varepsilon)\big) = I + \varepsilon(\Gamma^T + \Gamma) + o(\varepsilon).$$

Since $C(\varepsilon)^T C(\varepsilon) = I$, the coefficient of ε must vanish, hence

$$\Gamma^T + \Gamma = 0.$$

□

5.2 Symmetric and antisymmetric sectors

The infinitesimal generator admits a canonical decomposition into symmetric and antisymmetric parts:

$$\Gamma = S + A, \qquad S := \tfrac{1}{2}(\Gamma + \Gamma^T), \qquad A := \tfrac{1}{2}(\Gamma - \Gamma^T),$$

with

$$S^T = S, \qquad A^T = -A.$$

This decomposition separates two qualitatively different local behaviors. The symmetric sector S corresponds to local expansion or contraction, whereas the antisymmetric sector A generates local rotational behavior.

The stability condition (7) selects between these sectors sharply. Indeed, (7) is equivalent to

$$S = 0, \qquad \Gamma = A.$$

Thus stability acts as a sector-selection principle at the infinitesimal level: it excludes the symmetric local mode and retains only the antisymmetric one. The admissible stable local generator is therefore purely rotational in its canonical real local form.

5.3 Canonical local form

A real skew-symmetric matrix admits an orthogonal block decomposition into 2×2 rotation blocks and possibly zero blocks [1, 2]. Thus every real stable local generator decomposes into independent planar rotational modes, together with possible trivial directions. On each nontrivial invariant two-dimensional subspace one may choose an oriented orthonormal basis in which

$$A = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix} = \omega J, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

Hence the minimal nontrivial stable local block is rotational. Its finite local iterate is

$$C(\varepsilon) = \exp(\varepsilon A) = \exp(\varepsilon\omega J).$$

Using $J^2 = -I$, the exponential splits into even and odd powers:

$$\exp(\theta J) = \cos\theta\, I + \sin\theta\, J, \qquad \theta := \varepsilon\omega.$$

Thus the minimal nontrivial stable local mode is oscillatory.

Theorem 1 (Canonical stable local block)**.** *Let $C(\varepsilon)$ be an embeddable locally regular iteration family satisfying*

$$C(\varepsilon)^T C(\varepsilon) = I.$$

Then its induced local generator is skew-symmetric and decomposes into canonical 2×2 rotation blocks and possible zero blocks. In particular, the minimal nontrivial stable local mode is two-dimensional and oscillatory.

Proof. By Proposition 1, the induced local generator satisfies

$$\Gamma^T + \Gamma = 0,$$

so it is real skew-symmetric. By the standard orthogonal canonical form for real skew-symmetric matrices [1, 2], Γ decomposes into 2×2 rotation blocks and possible zero blocks. On each nontrivial block one may write

$$\Gamma = \omega J, \qquad J^2 = -I.$$

Exponentiation then gives

$$\exp(\varepsilon\Gamma) = \exp(\varepsilon\omega J) = \cos(\varepsilon\omega) I + \sin(\varepsilon\omega) J,$$

so the minimal nontrivial stable local mode is two-dimensional and oscillatory. □

5.4 Scalar no-go

We now specialize to the one-dimensional real scalar sector. In that case,

$$C(\varepsilon) = c(\varepsilon) \in \mathbb{R},$$

and the stability condition becomes simply

$$c(\varepsilon)^2 = 1.$$

If the iteration family is continuous at $\varepsilon = 0$ and satisfies $c(0) = 1$, then necessarily

$$c(\varepsilon) \equiv 1$$

in a neighborhood of $\varepsilon = 0$. Equivalently, if one writes

$$c(\varepsilon) = \exp(\varepsilon A), \qquad A \in \mathbb{R},$$

then

$$\exp(2\varepsilon A) = 1$$

for all sufficiently small ε, which implies $A = 0$.

Theorem 2 (Real scalar no-go)**.** *The real one-dimensional stable local sector is necessarily trivial. In particular, there exists no nontrivial real scalar local generator compatible with*

$$C(\varepsilon)^T C(\varepsilon) = I.$$

Proof. In the one-dimensional real sector, write $C(\varepsilon) = c(\varepsilon) \in \mathbb{R}$. The stability condition becomes

$$c(\varepsilon)^2 = 1.$$

If $c(0) = 1$ and $c(\varepsilon)$ is continuous at $\varepsilon = 0$, then $c(\varepsilon)$ cannot jump to -1 in a neighborhood of 0. Hence

$$c(\varepsilon) \equiv 1$$

near $\varepsilon = 0$. Equivalently, if $c(\varepsilon) = \exp(\varepsilon A)$ with $A \in \mathbb{R}$, then

$$\exp(2\varepsilon A) = 1$$

for all sufficiently small ε, which implies $A = 0$. Therefore no nontrivial real scalar local generator exists. □

The only admissible real scalar generator is therefore $A = 0$, which yields the identity family $C(\varepsilon) \equiv I$. Hence the real one-dimensional stable local sector admits only the trivial identity family and therefore carries no nontrivial local dynamics. The first nontrivial stable local mode appears only in the canonical real 2×2 rotation block.

Any locally iterable family must approach the identity as $\varepsilon \to 0$. In a scalar representation, a nonzero real generator produces local amplification or local collapse, while the zero generator yields only the trivial identity family. Hence the minimal nontrivial stable local behavior compatible with this identity limit must be oscillatory.

5.5 Complex scalar representation of the canonical block

The scalar no-go theorem shows that in the real one-dimensional case, skew-symmetry forces

$$A = 0.$$

Thus the real 1×1 sector admits no nontrivial stable local generator. The first nontrivial stable local sector therefore appears in dimension two, where the canonical antisymmetric block has the form

$$A = \omega J, \qquad J^2 = -I.$$

The relation $J^2 = -I$ has a precise algebraic meaning: it endows the underlying real two-dimensional invariant subspace with a complex structure.

Lemma 1 (Complex structure induced by the canonical block)**.** *Let V be a real two-dimensional vector space, and let*

$$J : V \to V$$

be a linear operator satisfying

$$J^2 = -I.$$

Then J defines on V the structure of a one-dimensional complex vector space by

$$(a + ib) \cdot v := a\,v + b\,Jv, \qquad a, b \in \mathbb{R}, \quad v \in V.$$

With respect to this complex structure, the operator J acts as multiplication by i.

Proof. Define scalar multiplication by

$$(a + ib) \cdot v := a\,v + b\,Jv.$$

It is enough to verify compatibility with complex multiplication. Let

$$z_1 = a + ib, \qquad z_2 = c + id.$$

Then

$$z_1 \cdot (z_2 \cdot v) = (a + ib) \cdot (cv + dJv).$$

Using linearity of J and the identity $J^2 = -I$, we obtain

$$\begin{aligned} z_1 \cdot (z_2 \cdot v) &= a(cv + dJv) + bJ(cv + dJv) \\ &= ac\,v + ad\,Jv + bc\,Jv + bd\,J^2 v \\ &= (ac - bd)v + (ad + bc)Jv. \end{aligned}$$

But

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc),$$

hence

$$z_1 \cdot (z_2 \cdot v) = \big((a + ib)(c + id)\big) \cdot v.$$

Therefore the above rule defines a complex vector space structure on V. By construction,

$$i \cdot v = Jv,$$

so J acts exactly as multiplication by i. □

Thus the canonical real two-dimensional stable block is not merely analogous to complex multiplication: it canonically induces it. In this sense, the first nontrivial stable local mode admits a natural one-dimensional complex representation. The appearance of complex scalar structure is therefore not an independent algebraic enrichment; it is the reduced representation of the minimal nontrivial stable real block.

Applying Lemma 1 to the canonical block

$$A = \omega J,$$

one obtains, in the associated one-dimensional complex representation,

$$A = \omega i.$$

Accordingly, the local iterate takes the scalar complex form

$$C(\varepsilon) = \exp(\varepsilon A) = \exp(i\varepsilon\omega),$$

and therefore

$$\exp(i\theta) = \cos\theta + i\sin\theta.$$

Corollary 3 (Emergence of the imaginary unit). The real one-dimensional stable local sector is necessarily trivial. The first nontrivial stable local mode appears in the canonical real two-dimensional rotation block. That block canonically defines a complex structure. In the associated one-dimensional complex representation, the canonical generator J is represented by multiplication by i, where

$$i^2 = -1.$$

Consequently, the full local generator $A = \omega J$ is represented by multiplication by $i\omega$ for some real ω. In this sense, the imaginary unit is not postulated independently, but appears as the scalar realization of the minimal nontrivial stable local mode.

Theorem 4 (Minimal stable local dynamics is canonically complex). *Let $C(\varepsilon)$ be a nontrivial embeddable locally regular iteration family satisfying*

$$C(\varepsilon)^T C(\varepsilon) = I.$$

Then its minimal nontrivial stable local realization cannot be real one-dimensional. Its first admissible nontrivial stable local realization is necessarily two-dimensional, rotational, and carries a natural complex structure. In the associated one-dimensional complex representation, the generator is represented by multiplication by $i\omega$ for some real ω.

Proof. By Proposition 1, the induced local generator is skew-symmetric. By Theorem 2, no nontrivial real one-dimensional stable local generator exists. Hence the minimal nontrivial stable local realization cannot be real one-dimensional. By the canonical block decomposition of Theorem 1, the first nontrivial admissible block is two-dimensional and rotational, of the form

$$A = \omega J, \qquad J^2 = -I.$$

By Lemma 1, this block carries a natural complex structure, and in the associated one-dimensional complex representation one has

$$A = \omega i.$$

Therefore the first admissible nontrivial stable local realization is canonically complex. □

6 Conclusion

We have proposed a minimal criterion for when a relation can exist as a dynamical object. Dynamics should not be identified with the mere existence of an update rule. A relation qualifies as dynamical only if it admits coherent admissible self-continuation as one and the same relational object.

This requirement was formulated operationally through embeddability. A candidate relation is dynamical only if it can be embedded into a compositionally consistent iteration family with identity, closure under composition, and decomposition invariance. In this framework, local generators are not postulated at the outset. They arise only as derived local representations of an already existing iterable relation. In particular, the local differential law does not generate the dynamics; it records the local form of an already existing iterable structure.

To distinguish mere iterability from viable local dynamics, we imposed the minimal stability condition

$$C(\varepsilon)^T C(\varepsilon) = I.$$

Its infinitesimal consequence forces the induced local generator into the skew-symmetric sector. As a result, the minimal nontrivial stable local mode is oscillatory and is represented canonically by two-dimensional rotation blocks generated by an operator J satisfying

$$J^2 = -I.$$

This leads to a sharp structural consequence. Stable local dynamics cannot begin in the real scalar sector. The first admissible nontrivial stable local mode is necessarily two-dimensional and rotational. That mode carries a natural complex structure, and in the associated one-dimensional complex representation its generator is represented by multiplication by $i\omega$ for some real ω, where $i^2 = -1$. In this sense, complex local dynamics is not an additional postulate, but a necessary consequence of stable relational iterability.

The resulting picture is therefore both principle-based and compositional. Dynamics is primary not as a parameterized differential law, but as the existence of a coherent iterable relational object. Generators, local equations, and complex scalar realizations arise only as secondary representations of that underlying iterable structure.

References

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